Supplementary Material for the revision

Does the Choice of Metaheuristic Optimizer Matter? A Matched-Budget Benchmark for XGBoost Tuning in Molecular Property Prediction

Tables S1–S10 and Figures S1–S6 are unchanged from the original Supplementary Material, with one correction to Table S2: for BBBP in Experiment B the Friedman statistic is $\chi^2 = 10.96$, $p = 0.027$ (previously 11.44, 0.022). The earlier value came from test_summary.csv, whose 16-digit rounding broke exact ties between ROC-AUC values; the corrected value uses the exact per-run results. No test verdict changes. Tables S11–S21 report the analyses added in the revision, Supplementary Note S1 (at the end) describes their designs and detailed results, and Figures S7–S8 show the pre-refit diagnostic and the validation control. Every table is generated from the CSV file named in its note (folder analysis/revision/out of release v1.1.0).

Table S11. Paired optimizer differences in the primary test metric (goodness: ROC-AUC, or minus RMSE) and TOST equivalence.

Dataset	Exp.	Pair	Margin	Mean difference [95% CI]	Min. equiv. margin	TOST p (Holm)	Equivalent
ESOL	A	Random vs TPE	0.0178	-0.0025 [-0.0134, 0.0084]	0.0115	0.030	yes
ESOL	A	Random vs GWO	0.0178	-0.0097 [-0.0206, 0.0011]	0.0187	0.232	no
ESOL	A	Random vs WOA	0.0178	-0.0111 [-0.0237, 0.0016]	0.0216	0.232	no
ESOL	A	Random vs PSO	0.0178	-0.0074 [-0.0181, 0.0032]	0.0163	0.141	no
ESOL	A	TPE vs GWO	0.0178	-0.0073 [-0.0205, 0.0060]	0.0183	0.232	no
ESOL	A	TPE vs WOA	0.0178	-0.0086 [-0.0221, 0.0050]	0.0198	0.232	no
ESOL	A	TPE vs PSO	0.0178	-0.0049 [-0.0170, 0.0072]	0.0150	0.115	no
ESOL	A	GWO vs WOA	0.0178	-0.0013 [-0.0123, 0.0097]	0.0104	0.021	yes
ESOL	A	GWO vs PSO	0.0178	0.0023 [-0.0064, 0.0111]	0.0096	0.006	yes
ESOL	A	WOA vs PSO	0.0178	0.0036 [-0.0070, 0.0143]	0.0125	0.038	yes
ESOL	B	Random vs TPE	0.0178	-0.0352 [-0.0659, -0.0045]	0.0607	1.000	no
ESOL	B	Random vs GWO	0.0178	-0.0097 [-0.0436, 0.0243]	0.0378	1.000	no
ESOL	B	Random vs WOA	0.0178	-0.0087 [-0.0461, 0.0288]	0.0398	1.000	no
ESOL	B	Random vs PSO	0.0178	-0.0234 [-0.0545, 0.0077]	0.0492	1.000	no
ESOL	B	TPE vs GWO	0.0178	0.0255 [0.0028, 0.0482]	0.0444	1.000	no
ESOL	B	TPE vs WOA	0.0178	0.0265 [0.0062, 0.0468]	0.0434	1.000	no
ESOL	B	TPE vs PSO	0.0178	0.0118 [-0.0081, 0.0316]	0.0282	1.000	no
ESOL	B	GWO vs WOA	0.0178	0.0010 [-0.0278, 0.0298]	0.0249	1.000	no
ESOL	B	GWO vs PSO	0.0178	-0.0137 [-0.0357, 0.0082]	0.0320	1.000	no
ESOL	B	WOA vs PSO	0.0178	-0.0147 [-0.0394, 0.0099]	0.0352	1.000	no

FreeSolv	A	Random vs TPE	0.0562	-0.0603 [-0.1636, 0.0430]	0.1462	1.000	no
FreeSolv	A	Random vs GWO	0.0562	-0.1393 [-0.2295, -0.0492]	0.2142	1.000	no
FreeSolv	A	Random vs WOA	0.0562	-0.0175 [-0.1276, 0.0926]	0.1090	1.000	no
FreeSolv	A	Random vs PSO	0.0562	-0.1068 [-0.2229, 0.0093]	0.2032	1.000	no
FreeSolv	A	TPE vs GWO	0.0562	-0.0790 [-0.1644, 0.0063]	0.1500	1.000	no
FreeSolv	A	TPE vs WOA	0.0562	0.0428 [-0.0424, 0.1280]	0.1136	1.000	no
FreeSolv	A	TPE vs PSO	0.0562	-0.0465 [-0.1218, 0.0288]	0.1090	1.000	no
FreeSolv	A	GWO vs WOA	0.0562	0.1218 [0.0518, 0.1919]	0.1800	1.000	no
FreeSolv	A	GWO vs PSO	0.0562	0.0326 [-0.0583, 0.1234]	0.1080	1.000	no
FreeSolv	A	WOA vs PSO	0.0562	-0.0893 [-0.1781, -0.0004]	0.1631	1.000	no
FreeSolv	B	Random vs TPE	0.0562	0.0227 [-0.0786, 0.1240]	0.1068	1.000	no
FreeSolv	B	Random vs GWO	0.0562	-0.0198 [-0.1353, 0.0956]	0.1157	1.000	no
FreeSolv	B	Random vs WOA	0.0562	0.0660 [-0.0585, 0.1906]	0.1695	1.000	no
FreeSolv	B	Random vs PSO	0.0562	-0.0290 [-0.1127, 0.0547]	0.0986	1.000	no
FreeSolv	B	TPE vs GWO	0.0562	-0.0425 [-0.1512, 0.0661]	0.1328	1.000	no
FreeSolv	B	TPE vs WOA	0.0562	0.0433 [-0.0858, 0.1725]	0.1506	1.000	no
FreeSolv	B	TPE vs PSO	0.0562	-0.0517 [-0.1527, 0.0492]	0.1356	1.000	no
FreeSolv	B	GWO vs WOA	0.0562	0.0859 [-0.0253, 0.1971]	0.1782	1.000	no
FreeSolv	B	GWO vs PSO	0.0562	-0.0092 [-0.1253, 0.1069]	0.1057	1.000	no
FreeSolv	B	WOA vs PSO	0.0562	-0.0951 [-0.2046, 0.0144]	0.1861	1.000	no
Lipophilicity	A	Random vs TPE	0.0152	-0.0065 [-0.0125, -0.0006]	0.0115	0.003	yes
Lipophilicity	A	Random vs GWO	0.0152	-0.0054 [-0.0110, 0.0002]	0.0101	0.002	yes
Lipophilicity	A	Random vs WOA	0.0152	-0.0025 [-0.0086, 0.0036]	0.0076	<0.001	yes
Lipophilicity	A	Random vs PSO	0.0152	-0.0025 [-0.0088, 0.0039]	0.0077	<0.001	yes
Lipophilicity	A	TPE vs GWO	0.0152	0.0011 [-0.0037, 0.0060]	0.0052	<0.001	yes
Lipophilicity	A	TPE vs WOA	0.0152	0.0040 [-0.0028, 0.0109]	0.0097	0.002	yes
Lipophilicity	A	TPE vs PSO	0.0152	0.0041 [-0.0021, 0.0103]	0.0092	0.002	yes
Lipophilicity	A	GWO vs WOA	0.0152	0.0029 [-0.0036, 0.0094]	0.0083	0.002	yes
Lipophilicity	A	GWO vs PSO	0.0152	0.0030 [-0.0028, 0.0087]	0.0078	<0.001	yes
Lipophilicity	A	WOA vs PSO	0.0152	0.0001 [-0.0057, 0.0058]	0.0048	<0.001	yes

Lipophilicity	B	Random vs TPE	0.0152	0.0014 [-0.0047, 0.0076]	0.0065	<0.001	yes
Lipophilicity	B	Random vs GWO	0.0152	-0.0029 [-0.0102, 0.0044]	0.0090	0.004	yes
Lipophilicity	B	Random vs WOA	0.0152	0.0007 [-0.0065, 0.0079]	0.0067	<0.001	yes
Lipophilicity	B	Random vs PSO	0.0152	-0.0017 [-0.0095, 0.0061]	0.0082	0.003	yes
Lipophilicity	B	TPE vs GWO	0.0152	-0.0043 [-0.0115, 0.0028]	0.0103	0.004	yes
Lipophilicity	B	TPE vs WOA	0.0152	-0.0007 [-0.0075, 0.0061]	0.0064	<0.001	yes
Lipophilicity	B	TPE vs PSO	0.0152	-0.0031 [-0.0107, 0.0044]	0.0094	0.004	yes
Lipophilicity	B	GWO vs WOA	0.0152	0.0036 [-0.0033, 0.0105]	0.0093	0.004	yes
Lipophilicity	B	GWO vs PSO	0.0152	0.0012 [-0.0051, 0.0075]	0.0064	<0.001	yes
Lipophilicity	B	WOA vs PSO	0.0152	-0.0024 [-0.0085, 0.0037]	0.0075	<0.001	yes
BBBP	A	Random vs TPE	0.0100	0.0010 [-0.0069, 0.0089]	0.0076	0.082	no
BBBP	A	Random vs GWO	0.0100	-0.0086 [-0.0144, -0.0028]	0.0134	1.000	no
BBBP	A	Random vs WOA	0.0100	-0.0004 [-0.0068, 0.0060]	0.0057	0.022	yes
BBBP	A	Random vs PSO	0.0100	0.0000 [-0.0068, 0.0068]	0.0056	0.022	yes
BBBP	A	TPE vs GWO	0.0100	-0.0096 [-0.0167, -0.0024]	0.0155	1.000	no
BBBP	A	TPE vs WOA	0.0100	-0.0014 [-0.0096, 0.0068]	0.0082	0.099	no
BBBP	A	TPE vs PSO	0.0100	-0.0010 [-0.0086, 0.0066]	0.0073	0.075	no
BBBP	A	GWO vs WOA	0.0100	0.0082 [0.0019, 0.0144]	0.0134	1.000	no
BBBP	A	GWO vs PSO	0.0100	0.0086 [0.0019, 0.0152]	0.0141	1.000	no
BBBP	A	WOA vs PSO	0.0100	0.0004 [-0.0059, 0.0067]	0.0057	0.022	yes
BBBP	B	Random vs TPE	0.0100	-0.0062 [-0.0124, 0.0000]	0.0113	0.539	no
BBBP	B	Random vs GWO	0.0100	-0.0044 [-0.0107, 0.0019]	0.0096	0.277	no
BBBP	B	Random vs WOA	0.0100	0.0033 [-0.0042, 0.0108]	0.0096	0.277	no
BBBP	B	Random vs PSO	0.0100	-0.0058 [-0.0135, 0.0018]	0.0122	0.557	no
BBBP	B	TPE vs GWO	0.0100	0.0018 [-0.0039, 0.0074]	0.0065	0.029	yes
BBBP	B	TPE vs WOA	0.0100	0.0095 [0.0030, 0.0160]	0.0149	0.834	no
BBBP	B	TPE vs PSO	0.0100	0.0003 [-0.0069, 0.0075]	0.0063	0.045	yes
BBBP	B	GWO vs WOA	0.0100	0.0077 [0.0006, 0.0148]	0.0136	0.775	no
BBBP	B	GWO vs PSO	0.0100	-0.0014 [-0.0091, 0.0062]	0.0078	0.115	no
BBBP	B	WOA vs PSO	0.0100	-0.0092 [-0.0173, -0.0010]	0.0159	0.834	no

BACE	A	Random vs TPE	0.0100	0.0010 [-0.0069, 0.0089]	0.0075	0.119	no
BACE	A	Random vs GWO	0.0100	-0.0048 [-0.0094, -0.0002]	0.0086	0.119	no
BACE	A	Random vs WOA	0.0100	-0.0014 [-0.0101, 0.0072]	0.0086	0.181	no
BACE	A	Random vs PSO	0.0100	-0.0057 [-0.0118, 0.0004]	0.0108	0.339	no
BACE	A	TPE vs GWO	0.0100	-0.0058 [-0.0138, 0.0022]	0.0124	0.339	no
BACE	A	TPE vs WOA	0.0100	-0.0024 [-0.0125, 0.0077]	0.0108	0.339	no
BACE	A	TPE vs PSO	0.0100	-0.0067 [-0.0143, 0.0009]	0.0130	0.339	no
BACE	A	GWO vs WOA	0.0100	0.0033 [-0.0047, 0.0114]	0.0100	0.305	no
BACE	A	GWO vs PSO	0.0100	-0.0009 [-0.0063, 0.0044]	0.0054	0.009	yes
BACE	A	WOA vs PSO	0.0100	-0.0043 [-0.0129, 0.0043]	0.0114	0.339	no
BACE	B	Random vs TPE	0.0100	-0.0012 [-0.0104, 0.0080]	0.0088	0.179	no
BACE	B	Random vs GWO	0.0100	-0.0028 [-0.0117, 0.0061]	0.0102	0.272	no
BACE	B	Random vs WOA	0.0100	-0.0015 [-0.0091, 0.0061]	0.0078	0.118	no
BACE	B	Random vs PSO	0.0100	-0.0076 [-0.0149, -0.0002]	0.0137	0.408	no
BACE	B	TPE vs GWO	0.0100	-0.0016 [-0.0087, 0.0055]	0.0075	0.102	no
BACE	B	TPE vs WOA	0.0100	-0.0003 [-0.0084, 0.0078]	0.0070	0.102	no
BACE	B	TPE vs PSO	0.0100	-0.0064 [-0.0152, 0.0025]	0.0137	0.408	no
BACE	B	GWO vs WOA	0.0100	0.0013 [-0.0067, 0.0093]	0.0080	0.121	no
BACE	B	GWO vs PSO	0.0100	-0.0047 [-0.0128, 0.0033]	0.0115	0.334	no
BACE	B	WOA vs PSO	0.0100	-0.0061 [-0.0117, -0.0004]	0.0108	0.334	no

Margins: 2% of the default test RMSE (regression) or 0.01 ROC-AUC. Minimal equivalence margin = largest absolute bound of the 90% CI. Holm correction over the 10 pairs of a cell. Source: R1_4_pairwise_equivalence.csv.

Table S12a. Model-seed control: spread of the test score of each selected configuration over three XGBoost random states.

Dataset	Exp.	Method	SD over model seeds	SD over tuning seeds	Model-seed share
BACE	A	GWO	0.0097	0.0119	0.61
BACE	A	PSO	0.0110	0.0148	0.65
BACE	A	Random	0.0101	0.0122	0.61
BACE	A	TPE	0.0149	0.0196	0.64
BACE	A	WOA	0.0102	0.0163	0.49
BACE	B	GWO	0.0102	0.0140	0.36
BACE	B	PSO	0.0073	0.0150	0.23
BACE	B	Random	0.0104	0.0158	0.42
BACE	B	TPE	0.0094	0.0161	0.44
BACE	B	WOA	0.0087	0.0160	0.24
BBBP	A	GWO	0.0076	0.0110	0.58
BBBP	A	PSO	0.0101	0.0131	0.53

BBBP	A	Random	0.0101	0.0121	0.62
BBBP	A	TPE	0.0116	0.0161	0.49
BBBP	A	WOA	0.0077	0.0143	0.28
BBBP	B	GWO	0.0073	0.0126	0.38
BBBP	B	PSO	0.0081	0.0140	0.21
BBBP	B	Random	0.0115	0.0147	0.51
BBBP	B	TPE	0.0068	0.0113	0.29
BBBP	B	WOA	0.0078	0.0160	0.24
ESOL	A	GWO	0.0177	0.0192	0.66
ESOL	A	PSO	0.0192	0.0231	0.58
ESOL	A	Random	0.0164	0.0211	0.39
ESOL	A	TPE	0.0163	0.0255	0.41
ESOL	A	WOA	0.0167	0.0271	0.38
ESOL	B	GWO	0.0172	0.0532	0.13
ESOL	B	PSO	0.0196	0.0351	0.32
ESOL	B	Random	0.0240	0.0758	0.11
ESOL	B	TPE	0.0199	0.0284	0.33
ESOL	B	WOA	0.0197	0.0538	0.14
FreeSolv	A	GWO	0.1403	0.1805	0.71
FreeSolv	A	PSO	0.1136	0.1364	0.61
FreeSolv	A	Random	0.1363	0.2368	0.37
FreeSolv	A	TPE	0.1224	0.1734	0.46
FreeSolv	A	WOA	0.1116	0.1904	0.54
FreeSolv	B	GWO	0.1321	0.2492	0.30
FreeSolv	B	PSO	0.1062	0.1826	0.34
FreeSolv	B	Random	0.0972	0.2133	0.20
FreeSolv	B	TPE	0.1075	0.2102	0.24
FreeSolv	B	WOA	0.0973	0.2505	0.13
Lipophilicity	A	GWO	0.0065	0.0088	0.55
Lipophilicity	A	PSO	0.0071	0.0134	0.25
Lipophilicity	A	Random	0.0075	0.0130	0.39
Lipophilicity	A	TPE	0.0067	0.0105	0.34
Lipophilicity	A	WOA	0.0070	0.0156	0.20
Lipophilicity	B	GWO	0.0077	0.0132	0.33
Lipophilicity	B	PSO	0.0078	0.0150	0.32
Lipophilicity	B	Random	0.0067	0.0108	0.28
Lipophilicity	B	TPE	0.0068	0.0135	0.23
Lipophilicity	B	WOA	0.0076	0.0163	0.20

Model-seed share = mean within-run variance over the three refits / total variance of the 90 values. Source: J1_model_seed.csv.

Table S12b. Optimizer comparison with the test score averaged over three model seeds.

Dataset	Exp.	Friedman p (one refit)	Holm pairs (one refit)	Friedman p (mean of 3 refits)	Holm pairs (mean of 3 refits)
BACE	A	0.225	0	0.675	0
BACE	B	0.181	0	0.272	0
BBBP	A	0.061	0	0.002	3
BBBP	B	0.027	0	0.041	0
ESOL	A	0.106	0	0.332	0
ESOL	B	0.171	0	0.076	0
FreeSolv	A	0.009	2	<0.001	1
FreeSolv	B	0.495	0	0.697	0
Lipophilicity	A	0.247	0	0.184	0
Lipophilicity	B	0.512	0	0.890	0

Source: J1_model_seed_avg_comparison.csv.

Table S13a. Five independent scaffold partitions (Experiment A): optimizer effect and variance components.

Dataset	F (optimizer)	p	Var. partition	Var. optimizer × partition	Var. tuning seed	Default (mean)
ESOL	0.87	0.506	0.01290	0.00000	0.00214	0.950
FreeSolv	0.65	0.635	0.29971	0.00000	0.05881	2.396
BBBP	1.84	0.170	0.00098	0.00000	0.00042	0.915
BACE	1.09	0.395	0.00074	0.00006	0.00051	0.876

Optimizer effect tested against the optimizer × partition interaction (4 and 16 df). Source: J6_variance_components.csv.

Table S13b. Pairwise optimizer differences on partition means (goodness units).

Dataset	Pair	Mean difference [95% CI]	p (Holm)	Min. equiv. margin	Equivalent
ESOL	Random vs TPE	-0.0138 [-0.0326, 0.0051]	1.000	0.0283	no
ESOL	Random vs GWO	-0.0085 [-0.0416, 0.0246]	1.000	0.0339	no
ESOL	Random vs WOA	-0.0051 [-0.0438, 0.0337]	1.000	0.0348	no
ESOL	Random vs PSO	0.0073 [-0.0069, 0.0215]	1.000	0.0182	no
ESOL	TPE vs GWO	0.0052 [-0.0421, 0.0526]	1.000	0.0416	no
ESOL	TPE vs WOA	0.0087 [-0.0434, 0.0608]	1.000	0.0487	no
ESOL	TPE vs PSO	0.0211 [-0.0100, 0.0522]	1.000	0.0449	no
ESOL	GWO vs WOA	0.0035 [-0.0106, 0.0175]	1.000	0.0142	no
ESOL	GWO vs PSO	0.0158 [-0.0133, 0.0449]	1.000	0.0382	no
ESOL	WOA vs PSO	0.0124 [-0.0260, 0.0507]	1.000	0.0418	no
FreeSolv	Random vs TPE	-0.1139 [-0.3847, 0.1569]	1.000	0.3219	no
FreeSolv	Random vs GWO	-0.0144 [-0.1521, 0.1234]	1.000	0.1201	no
FreeSolv	Random vs WOA	-0.0836 [-0.3199, 0.1527]	1.000	0.2650	no
FreeSolv	Random vs PSO	-0.0377 [-0.1290, 0.0536]	1.000	0.1078	no
FreeSolv	TPE vs GWO	0.0996 [-0.2071, 0.4062]	1.000	0.3350	no
FreeSolv	TPE vs WOA	0.0303 [-0.3013, 0.3620]	1.000	0.2850	no
FreeSolv	TPE vs PSO	0.0762 [-0.2249, 0.3772]	1.000	0.3073	no
FreeSolv	GWO vs WOA	-0.0692 [-0.2433, 0.1049]	1.000	0.2029	no
FreeSolv	GWO vs PSO	-0.0234 [-0.1811, 0.1343]	1.000	0.1445	no
FreeSolv	WOA vs PSO	0.0458 [-0.1413, 0.2329]	1.000	0.1895	no
BBBP	Random vs TPE	-0.0052 [-0.0270, 0.0166]	1.000	0.0219	no
BBBP	Random vs GWO	-0.0025 [-0.0197, 0.0147]	1.000	0.0157	no
BBBP	Random vs WOA	-0.0143 [-0.0436, 0.0149]	1.000	0.0368	no
BBBP	Random vs PSO	-0.0116 [-0.0337, 0.0106]	1.000	0.0285	no
BBBP	TPE vs GWO	0.0027 [-0.0119, 0.0172]	1.000	0.0138	no
BBBP	TPE vs WOA	-0.0092 [-0.0231, 0.0048]	0.991	0.0198	no
BBBP	TPE vs PSO	-0.0064 [-0.0147, 0.0020]	0.910	0.0128	no
BBBP	GWO vs WOA	-0.0118 [-0.0285, 0.0049]	0.968	0.0247	no
BBBP	GWO vs PSO	-0.0090 [-0.0188, 0.0007]	0.613	0.0165	no
BBBP	WOA vs PSO	0.0028 [-0.0072, 0.0128]	1.000	0.0105	no

BACE	Random vs TPE	-0.0106 [-0.0490, 0.0278]	1.000	0.0400	no
BACE	Random vs GWO	-0.0191 [-0.0620, 0.0237]	1.000	0.0521	no
BACE	Random vs WOA	-0.0138 [-0.0515, 0.0240]	1.000	0.0427	no
BACE	Random vs PSO	-0.0131 [-0.0454, 0.0191]	1.000	0.0379	no
BACE	TPE vs GWO	-0.0086 [-0.0217, 0.0046]	1.000	0.0187	no
BACE	TPE vs WOA	-0.0032 [-0.0121, 0.0057]	1.000	0.0100	no
BACE	TPE vs PSO	-0.0026 [-0.0138, 0.0087]	1.000	0.0112	no
BACE	GWO vs WOA	0.0054 [-0.0126, 0.0234]	1.000	0.0192	no
BACE	GWO vs PSO	0.0060 [-0.0161, 0.0282]	1.000	0.0230	no
BACE	WOA vs PSO	0.0006 [-0.0084, 0.0097]	1.000	0.0076	no

t tests with 4 df. Source: J6_pairwise.csv.

Table S13c. Improvement of each optimizer over the default per partition (mean over three seeds; positive = better).

Dataset	Method	P1	P2	P3	P4	P5	Partitions better	Mean
ESOL	Random	+0.094	+0.016	+0.054	+0.028	+0.136	5	+0.066
ESOL	TPE	+0.121	+0.017	+0.088	+0.033	+0.138	5	+0.079
ESOL	GWO	+0.088	+0.008	+0.041	+0.048	+0.186	5	+0.074
ESOL	WOA	+0.090	+0.008	+0.033	+0.028	+0.195	5	+0.071
ESOL	PSO	+0.076	+0.012	+0.037	+0.037	+0.130	5	+0.058
FreeSolv	Random	+0.132	+0.712	+0.264	+0.066	-0.097	4	+0.215
FreeSolv	TPE	+0.609	+0.742	+0.155	+0.162	-0.023	4	+0.329
FreeSolv	GWO	+0.180	+0.551	+0.407	+0.072	-0.061	4	+0.230
FreeSolv	WOA	+0.371	+0.663	+0.556	-0.095	+0.000	4	+0.299
FreeSolv	PSO	+0.172	+0.778	+0.348	-0.024	-0.009	3	+0.253
BBBP	Random	-0.000	-0.025	-0.050	-0.007	-0.015	0	-0.019
BBBP	TPE	+0.019	-0.041	-0.026	-0.018	-0.006	1	-0.014
BBBP	GWO	-0.003	-0.037	-0.031	-0.014	+0.001	1	-0.017
BBBP	WOA	+0.016	-0.030	+0.000	-0.016	+0.005	3	-0.005
BBBP	PSO	+0.016	-0.030	-0.017	-0.016	+0.008	2	-0.008
BACE	Random	-0.047	-0.009	-0.031	-0.020	-0.010	0	-0.023
BACE	TPE	+0.017	-0.014	-0.041	-0.008	-0.018	1	-0.013
BACE	GWO	+0.029	-0.021	-0.036	+0.004	+0.004	3	-0.004
BACE	WOA	+0.016	-0.013	-0.030	+0.003	-0.023	2	-0.009
BACE	PSO	+0.010	-0.006	-0.025	-0.008	-0.022	1	-0.010

Source: J6_tuning_vs_default.csv.

Table S14. Pre-refit diagnostic: validation-test agreement when one model scores both sets.

Dataset	Exp.	Method	ρ pre-refit [95% CI]	ρ refitted [95% CI]	Median gap pre-refit	Median gap refitted	Default gap pre-refit
BACE	A	GWO	-0.16 [-0.52, 0.26]	-0.16 [-0.54, 0.28]	-0.081	-0.070	-0.054
BACE	A	PSO	-0.13 [-0.52, 0.29]	-0.14 [-0.52, 0.27]	-0.081	-0.068	-0.054
BACE	A	Random	-0.40 [-0.66, -0.05]	-0.45 [-0.73, -0.09]	-0.086	-0.073	-0.054
BACE	A	TPE	0.05 [-0.36, 0.43]	0.09 [-0.29, 0.45]	-0.093	-0.074	-0.054
BACE	A	WOA	0.18 [-0.20, 0.53]	0.02 [-0.31, 0.36]	-0.086	-0.070	-0.054
BACE	B	GWO	0.26 [-0.13, 0.59]	-0.11 [-0.45, 0.24]	-0.092	-0.090	-0.054
BACE	B	PSO	0.08 [-0.32, 0.47]	0.12 [-0.27, 0.49]	-0.084	-0.075	-0.054
BACE	B	Random	0.07 [-0.36, 0.46]	-0.07 [-0.48, 0.38]	-0.084	-0.077	-0.054
BACE	B	TPE	0.22 [-0.21, 0.61]	0.01 [-0.39, 0.41]	-0.088	-0.083	-0.054
BACE	B	WOA	-0.03 [-0.44, 0.36]	-0.05 [-0.42, 0.32]	-0.090	-0.077	-0.054
BBBP	A	GWO	-0.22 [-0.56, 0.17]	-0.00 [-0.38, 0.38]	-0.040	-0.038	-0.004
BBBP	A	PSO	-0.13 [-0.53, 0.29]	-0.03 [-0.39, 0.35]	-0.049	-0.044	-0.004

BBBP	A	Random	-0.19 [-0.52, 0.20]	-0.42 [-0.67, -0.06]	-0.041	-0.040	-0.004
BBBP	A	TPE	-0.09 [-0.45, 0.28]	-0.28 [-0.63, 0.14]	-0.053	-0.047	-0.004
BBBP	A	WOA	-0.20 [-0.57, 0.20]	-0.20 [-0.57, 0.21]	-0.051	-0.046	-0.004
BBBP	B	GWO	-0.22 [-0.51, 0.13]	-0.09 [-0.45, 0.28]	-0.051	-0.046	-0.004
BBBP	B	PSO	0.06 [-0.30, 0.39]	0.05 [-0.29, 0.38]	-0.042	-0.040	-0.004
BBBP	B	Random	0.14 [-0.23, 0.49]	0.07 [-0.27, 0.39]	-0.044	-0.043	-0.004
BBBP	B	TPE	-0.04 [-0.43, 0.36]	0.17 [-0.25, 0.56]	-0.044	-0.042	-0.004
BBBP	B	WOA	-0.38 [-0.63, -0.01]	-0.24 [-0.59, 0.19]	-0.048	-0.046	-0.004
ESOL	A	GWO	-0.09 [-0.48, 0.32]	-0.11 [-0.52, 0.28]	+0.272	+0.207	+0.080
ESOL	A	PSO	-0.20 [-0.53, 0.19]	-0.16 [-0.50, 0.21]	+0.263	+0.195	+0.080
ESOL	A	Random	-0.37 [-0.64, -0.02]	-0.13 [-0.45, 0.21]	+0.249	+0.190	+0.080
ESOL	A	TPE	-0.19 [-0.60, 0.23]	-0.02 [-0.42, 0.37]	+0.261	+0.204	+0.080
ESOL	A	WOA	-0.09 [-0.45, 0.28]	0.08 [-0.31, 0.41]	+0.232	+0.181	+0.080
ESOL	B	GWO	0.09 [-0.29, 0.45]	-0.04 [-0.40, 0.36]	+0.258	+0.204	+0.080
ESOL	B	PSO	-0.14 [-0.51, 0.26]	-0.16 [-0.52, 0.23]	+0.236	+0.187	+0.080
ESOL	B	Random	-0.00 [-0.36, 0.34]	-0.17 [-0.54, 0.21]	+0.250	+0.182	+0.080
ESOL	B	TPE	0.19 [-0.17, 0.54]	0.19 [-0.22, 0.54]	+0.235	+0.182	+0.080
ESOL	B	WOA	0.00 [-0.41, 0.41]	0.05 [-0.33, 0.39]	+0.232	+0.188	+0.080
FreeSolv	A	GWO	0.49 [0.20, 0.69]	0.42 [0.07, 0.72]	+0.300	-0.077	-0.821
FreeSolv	A	PSO	0.11 [-0.30, 0.49]	0.13 [-0.27, 0.52]	+0.208	-0.134	-0.821
FreeSolv	A	Random	0.49 [0.17, 0.71]	0.10 [-0.33, 0.49]	+0.097	-0.091	-0.821
FreeSolv	A	TPE	0.47 [0.07, 0.76]	0.17 [-0.21, 0.48]	+0.277	+0.117	-0.821
FreeSolv	A	WOA	0.28 [-0.16, 0.65]	0.23 [-0.19, 0.62]	+0.378	+0.061	-0.821
FreeSolv	B	GWO	0.52 [0.17, 0.76]	0.28 [-0.10, 0.58]	+0.517	+0.372	-0.821
FreeSolv	B	PSO	0.24 [-0.17, 0.61]	-0.19 [-0.51, 0.19]	+0.375	+0.165	-0.821
FreeSolv	B	Random	0.30 [-0.06, 0.61]	0.05 [-0.31, 0.43]	+0.187	-0.003	-0.821
FreeSolv	B	TPE	0.42 [0.08, 0.68]	0.24 [-0.11, 0.54]	+0.413	+0.371	-0.821
FreeSolv	B	WOA	-0.03 [-0.42, 0.37]	-0.32 [-0.66, 0.08]	+0.375	+0.144	-0.821
Lipophilicity	A	GWO	0.21 [-0.11, 0.52]	-0.28 [-0.61, 0.14]	+0.045	+0.037	+0.063
Lipophilicity	A	PSO	0.31 [-0.06, 0.61]	0.26 [-0.15, 0.62]	+0.039	+0.035	+0.063
Lipophilicity	A	Random	-0.20 [-0.53, 0.18]	0.04 [-0.32, 0.37]	+0.040	+0.031	+0.063
Lipophilicity	A	TPE	0.09 [-0.33, 0.48]	-0.09 [-0.44, 0.29]	+0.038	+0.032	+0.063
Lipophilicity	A	WOA	-0.02 [-0.41, 0.37]	0.00 [-0.40, 0.38]	+0.039	+0.030	+0.063
Lipophilicity	B	GWO	0.38 [0.03, 0.66]	0.39 [0.03, 0.67]	+0.049	+0.038	+0.063
Lipophilicity	B	PSO	0.33 [-0.06, 0.64]	0.33 [-0.07, 0.65]	+0.047	+0.038	+0.063
Lipophilicity	B	Random	0.14 [-0.26, 0.50]	0.22 [-0.15, 0.53]	+0.034	+0.028	+0.063
Lipophilicity	B	TPE	0.15 [-0.19, 0.45]	0.05 [-0.31, 0.38]	+0.050	+0.043	+0.063
Lipophilicity	B	WOA	0.44 [0.05, 0.73]	0.32 [-0.13, 0.69]	+0.041	+0.035	+0.063

Gap = test minus validation in the primary metric (RMSE, or ROC-AUC). Source: J1_prerefit_diagnostic.csv.

Table S15. Validation scores of all searched configurations (median over runs of the per-run quantiles) and of the 150 selected configurations per cell.

Dataset	Exp.	Searched 5%	Searched median	Searched 95%	Selected min	Selected max	Selected range / searched 5–95% range
BACE	A	0.849	0.868	0.881	0.875	0.905	0.96

BACE	B	0.844	0.874	0.892	0.871	0.918	0.98
BBBP	A	0.917	0.933	0.941	0.938	0.958	0.84
BBBP	B	0.914	0.934	0.946	0.932	0.965	1.06
ESOL	A	0.735	0.766	0.829	0.689	0.754	0.69
ESOL	B	0.746	0.790	0.900	0.678	0.786	0.70
FreeSolv	A	2.703	2.928	3.332	1.829	2.814	1.57
FreeSolv	B	2.417	2.843	3.562	1.574	2.875	1.14
Lipophilicity	A	0.700	0.713	0.760	0.681	0.712	0.52
Lipophilicity	B	0.710	0.732	0.809	0.678	0.722	0.44

Primary validation metric (RMSE, or ROC-AUC). Source: R2_2_val_ranges.csv.

Table S16. Validation control on a pool of 200 random Experiment A configurations.

Quantity	ESOL	FreeSolv	Lipophilicity	BBBP	BACE
ρ(single, pre-refit test), 200 configs	0.03	-0.13	0.81	0.46	0.35
ρ(single, pre-refit test), best 20	0.06	0.07	-0.38	0.13	-0.39
P1 difference [95% CI]	-0.04 [-0.45, 0.67]	-0.20 [-0.75, 0.54]	1.18 [0.77, 1.60]	0.33 [-0.25, 0.73]	0.74 [0.28, 1.08]
ρ(single, refit test), 200 configs	0.12	0.06	0.74	0.38	0.37
ρ(CV, CV-refit test), 200 configs	0.54	0.32	0.80	0.42	0.48
P3 difference, 200 [95% CI]	0.42 [0.29, 0.56]	0.26 [0.12, 0.40]	0.05 [0.01, 0.10]	0.03 [-0.13, 0.20]	0.11 [-0.04, 0.25]
ρ(single, refit test), best 20	-0.19	0.20	0.04	-0.24	-0.15
ρ(CV, CV-refit test), best 20	0.06	0.05	-0.17	-0.13	0.05
Mean regret, single split	0.096	0.543	0.016	0.014	0.030
Mean regret, inner CV	0.038	0.312	0.019	0.020	0.032
Regret difference [95% CI]	0.058 [0.056, 0.060]	0.231 [0.223, 0.239]	-0.003 [-0.003, -0.003]	-0.006 [-0.006, -0.005]	-0.002 [-0.003, -0.002]
Searches beating default, single (%)	1.1	87.4	100.0	50.4	46.0
Searches beating default, CV (%)	87.5	100.0	100.0	12.5	31.1
Default test score	0.889	2.811	0.760	0.915	0.814

Regret in RMSE or 1 − ROC-AUC (lower = better). 2,000 simulated random searches of 100 configurations; bootstrap CIs. Source: J2_validation_control.csv.

Table S17a. Original mirrored-encoding check: paired changes, all coordinates mirrored (M) minus standard (S).

Dataset	Method	Outcome	Mean change [95% CI]	Margin	TOST p	Min. equiv. margin
BBBP	Random	selected fraction (M - S)	-0.0024 [-0.0189, +0.0141]	0.0300	<0.001	0.0161
BBBP	Random	test goodness (M - S)	-0.0012 [-0.0088, +0.0064]	0.0100	0.013	0.0075
BBBP	TPE	selected fraction (M - S)	-0.0060 [-0.0271, +0.0150]	0.0300	0.014	0.0235
BBBP	TPE	test goodness (M - S)	+0.0014 [-0.0047, +0.0074]	0.0100	0.003	0.0064
BBBP	GWO	selected fraction (M - S)	+0.1119 [+0.0914, +0.1325]	0.0300	1.000	0.1290
BBBP	GWO	test goodness (M - S)	-0.0021 [-0.0104, +0.0061]	0.0100	0.030	0.0090
BBBP	WOA	selected fraction (M - S)	-0.0500 [-0.1129, +0.0129]	0.0300	0.740	0.1022
BBBP	WOA	test goodness (M - S)	+0.0041 [-0.0037, +0.0118]	0.0100	0.065	0.0105
BBBP	PSO	selected fraction (M - S)	-0.0038 [-0.0211, +0.0135]	0.0300	0.002	0.0181
BBBP	PSO	test goodness (M - S)	+0.0012 [-0.0070, +0.0093]	0.0100	0.017	0.0079

FreeSolv	Random	selected fraction (M - S)	-0.0090 [-0.0306, +0.0126]	0.0300	0.028	0.0269
FreeSolv	Random	test goodness (M - S)	-0.0437 [-0.1687, +0.0813]	0.0562	0.420	0.1475
FreeSolv	TPE	selected fraction (M - S)	+0.0129 [-0.0082, +0.0339]	0.0300	0.053	0.0304
FreeSolv	TPE	test goodness (M - S)	-0.0032 [-0.1046, +0.0981]	0.0562	0.147	0.0874
FreeSolv	GWO	selected fraction (M - S)	+0.1164 [+0.0924, +0.1403]	0.0300	1.000	0.1363
FreeSolv	GWO	test goodness (M - S)	+0.0241 [-0.0962, +0.1444]	0.0562	0.294	0.1240
FreeSolv	WOA	selected fraction (M - S)	-0.0520 [-0.1944, +0.0903]	0.0300	0.623	0.1703
FreeSolv	WOA	test goodness (M - S)	+0.0400 [-0.0709, +0.1508]	0.0562	0.383	0.1321
FreeSolv	PSO	selected fraction (M - S)	-0.0064 [-0.0277, +0.0148]	0.0300	0.015	0.0241
FreeSolv	PSO	test goodness (M - S)	-0.0483 [-0.1529, +0.0564]	0.0562	0.439	0.1352

Test goodness: ROC-AUC, or minus RMSE. Source: R1_5_mirror_paired_changes.csv.

Table S17b. Descriptor-only mirror (D) compared with the standard (S) and fully mirrored (M) encodings: selected fraction and test goodness.

Dataset	Method	Median S	Median M	Median D	Seeds D > S	Holm p (D vs S)	D − M [95% CI]	TOST p (Holm)	Test D − S [95% CI]
BBBP	Random	0.514	0.505	0.490	12	0.354	-0.013 [-0.031, +0.005]	0.099	+0.0052 [-0.0021, +0.0126]
BBBP	TPE	0.490	0.502	0.486	10	0.100	-0.013 [-0.032, +0.006]	0.099	+0.0003 [-0.0054, +0.0059]
BBBP	GWO	0.440	0.552	0.560	30	<0.001	+0.004 [-0.015, +0.023]	0.021	-0.0005 [-0.0065, +0.0055]
BBBP	WOA	0.500	0.464	0.476	13	0.995	+0.015 [-0.072, +0.101]	0.361	+0.0004 [-0.0077, +0.0086]
BBBP	PSO	0.500	0.495	0.500	15	0.995	+0.004 [-0.012, +0.019]	0.005	-0.0086 [-0.0180, +0.0008]
FreeSolv	Random	0.515	0.503	0.515	15	1.000	+0.011 [-0.005, +0.028]	0.053	+0.0067 [-0.1045, +0.1178]
FreeSolv	TPE	0.485	0.500	0.485	15	1.000	-0.010 [-0.030, +0.010]	0.053	+0.0495 [-0.0681, +0.1670]
FreeSolv	GWO	0.430	0.553	0.553	29	<0.001	+0.004 [-0.019, +0.028]	0.053	+0.0159 [-0.1049, +0.1367]
FreeSolv	WOA	0.579	0.500	0.398	12	0.103	-0.095 [-0.194, +0.004]	0.906	+0.0414 [-0.1223, +0.2051]
FreeSolv	PSO	0.488	0.494	0.500	16	1.000	+0.010 [-0.005, +0.025]	0.025	+0.0062 [-0.0802, +0.0925]

TOST margin ±0.03 in fraction. Source: J3_descriptor_mirror.csv.

Table S18. Prediction-level stability: Pearson correlation of test predictions between the 30 runs of a method.

Dataset	Exp.	Method	Mean pairwise r	Min pairwise r
BACE	A	GWO	0.944	0.868
BACE	A	PSO	0.946	0.853
BACE	A	Random	0.916	0.824
BACE	A	TPE	0.912	0.794
BACE	A	WOA	0.928	0.814
BACE	B	GWO	0.897	0.757
BACE	B	PSO	0.921	0.820
BACE	B	Random	0.899	0.780
BACE	B	TPE	0.899	0.782
BACE	B	WOA	0.890	0.628
BBBP	A	GWO	0.960	0.900
BBBP	A	PSO	0.953	0.865
BBBP	A	Random	0.951	0.853

BBBP	A	TPE	0.954	0.907
BBBP	A	WOA	0.948	0.858
BBBP	B	GWO	0.935	0.847
BBBP	B	PSO	0.941	0.854
BBBP	B	Random	0.926	0.846
BBBP	B	TPE	0.928	0.818
BBBP	B	WOA	0.929	0.838
ESOL	A	GWO	0.993	0.981
ESOL	A	PSO	0.990	0.971
ESOL	A	Random	0.990	0.972
ESOL	A	TPE	0.992	0.979
ESOL	A	WOA	0.988	0.970
ESOL	B	GWO	0.982	0.943
ESOL	B	PSO	0.980	0.942
ESOL	B	Random	0.978	0.924
ESOL	B	TPE	0.985	0.967
ESOL	B	WOA	0.977	0.937
FreeSolv	A	GWO	0.973	0.930
FreeSolv	A	PSO	0.975	0.941
FreeSolv	A	Random	0.970	0.917
FreeSolv	A	TPE	0.971	0.909
FreeSolv	A	WOA	0.974	0.919
FreeSolv	B	GWO	0.958	0.872
FreeSolv	B	PSO	0.971	0.934
FreeSolv	B	Random	0.964	0.903
FreeSolv	B	TPE	0.963	0.903
FreeSolv	B	WOA	0.957	0.835
Lipophilicity	A	GWO	0.977	0.957
Lipophilicity	A	PSO	0.973	0.928
Lipophilicity	A	Random	0.972	0.936
Lipophilicity	A	TPE	0.978	0.956
Lipophilicity	A	WOA	0.972	0.914
Lipophilicity	B	GWO	0.951	0.918
Lipophilicity	B	PSO	0.954	0.911
Lipophilicity	B	Random	0.953	0.902
Lipophilicity	B	TPE	0.957	0.932
Lipophilicity	B	WOA	0.949	0.906

Source: J1_prediction_stability.csv.

Table S19a. Subset-size penalty sensitivity (Experiment B, seeds 0–9).

Dataset	Penalty	Friedman p	Holm pairs	Kendall τ	Rank Random	Rank TPE	Rank GWO	Rank WOA	Rank PSO	GWO fraction
BBBP	0.01 (reference)	0.147	0	1.00	2.9	2.6	2.2	3.8	3.5	0.417
BBBP	0	0.114	0	0.60	2.8	3.4	1.9	3.6	3.3	0.421
BBBP	1% of default	0.147	0	0.60	2.6	3.3	2.2	3.9	3.0	0.421
BBBP	5% of default	0.439	0	0.40	2.5	3.0	2.8	3.8	2.9	0.431
FreeSolv	0.01 (reference)	0.487	0	1.00	2.4	3.1	3.2	3.6	2.7	0.433
FreeSolv	0	0.499	0	0.80	2.5	3.1	3.0	3.7	2.7	0.433
FreeSolv	1% of default	0.428	0	0.95	2.4	3.2	3.2	3.6	2.6	0.433
FreeSolv	5% of default	0.374	0	1.00	2.5	2.8	3.2	3.8	2.7	0.424

Source: J4_penalty.csv.

Table S19b. Mirrored encoding without any penalty (seeds 0–9).

Dataset	Method	Median fraction, standard	Median fraction, mirrored	Seeds mirrored > standard	Wilcoxon p
BBBP	GWO	0.421	0.560	10	0.002
BBBP	Random	0.514	0.495	4	0.625
FreeSolv	GWO	0.433	0.520	9	0.010
FreeSolv	Random	0.529	0.506	5	0.477

Source: J4_mirror_lam0.csv.

Table S20. Budget of 300 versus 100 calls (Experiment B, seeds 0–9). Gains are positive when the 300-call run is better.

Dataset	Method	Validation gain [95% CI]	Holm p	Test gain [95% CI]	Holm p	Fraction 100 / 300	Stability 100 / 300	Compute ratio
BBBP	Random	+0.0016 [-0.0005, +0.0037]	0.250	-0.0011 [-0.0082, +0.0061]	1.000	0.507 / 0.490	0.012 / 0.007	2.81
BBBP	TPE	+0.0064 [+0.0040, +0.0087]	0.010	-0.0072 [-0.0177, +0.0033]	0.641	0.521 / 0.512	-0.001 / 0.014	3.76
BBBP	GWO	+0.0048 [+0.0007, +0.0090]	0.082	-0.0014 [-0.0171, +0.0142]	1.000	0.417 / 0.379	0.025 / 0.035	2.76
BBBP	WOA	+0.0013 [-0.0026, +0.0052]	0.432	-0.0096 [-0.0303, +0.0111]	0.967	0.562 / 0.557	-0.032 / -0.022	3.55
BBBP	PSO	+0.0051 [+0.0025, +0.0078]	0.016	+0.0111 [+0.0012, +0.0209]	0.215	0.500 / 0.500	0.012 / -0.003	3.20
FreeSolv	Random	+0.1289 [+0.0065, +0.2513]	0.062	-0.0165 [-0.1312, +0.0981]	1.000	0.529 / 0.535	0.025 / 0.024	2.75
FreeSolv	TPE	+0.1830 [+0.0908, +0.2752]	0.012	+0.0813 [-0.0315, +0.1940]	1.000	0.471 / 0.491	0.025 / 0.039	2.84
FreeSolv	GWO	+0.5620 [+0.3969, +0.7271]	0.010	+0.0464 [-0.2066, +0.2993]	1.000	0.433 / 0.363	0.005 / 0.057	2.08
FreeSolv	WOA	+0.1556 [-0.0538, +0.3649]	0.131	-0.0049 [-0.1458, +0.1360]	1.000	0.722 / 0.629	-0.039 / -0.017	2.24
FreeSolv	PSO	+0.2239 [+0.1088, +0.3391]	0.010	-0.0788 [-0.1615, +0.0039]	0.322	0.482 / 0.456	-0.005 / 0.007	2.75

Friedman p across optimizers at 300 calls: BBBP 0.107, FreeSolv 0.395. Source: J5_budget.csv.

Table S21a. Standardized parent structures (largest fragment, neutralized, canonical tautomer, no stereochemistry).

Dataset	Parent groups (>1 molecule)	Groups crossing splits	Label-conflict groups	Validation excluded	Test excluded
ESOL	13	0	1	0 / 113	0 / 113
FreeSolv	3	0	0	0 / 64	0 / 65
Lipophilicity	82	0	3	0 / 420	0 / 420
BBBP	92	3	15	2 / 204	1 / 204
BACE	45	12	18	3 / 151	9 / 152

Source: J1_parents_summary.csv.

Table S21b. Optimizer comparison and tuning verdicts before and after excluding cross-split parents and label conflicts.

Dataset	Exp.	Version	Friedman p	Holm pairs	Tuned vs default (R, T, G, W, P)
ESOL	A	original	0.106	0	-----
ESOL	A	cleaned	0.106	0	-----
ESOL	B	original	0.171	0	-----
ESOL	B	cleaned	0.171	0	-----
FreeSolv	A	original	0.009	2	+++++
FreeSolv	A	cleaned	0.009	2	+++++

FreeSolv	B	original	0.495	0	+++++
FreeSolv	B	cleaned	0.495	0	+++++
Lipophilicity	A	original	0.247	0	+++++
Lipophilicity	A	cleaned	0.247	0	+++++
Lipophilicity	B	original	0.512	0	+++++
Lipophilicity	B	cleaned	0.512	0	+++++
BBBP	A	original	0.061	0	-----
BBBP	A	cleaned	0.058	0	-----
BBBP	B	original	0.027	0	-----
BBBP	B	cleaned	0.027	0	-----
BACE	A	original	0.225	0	00000
BACE	A	cleaned	0.344	0	00000
BACE	B	original	0.181	0	00000
BACE	B	cleaned	0.047	0	0000+

+ better than default, − worse, 0 no difference (Holm, $p < 0.05$). Selected configurations fixed; refitted test predictions re-scored. Source: J1_parents_conclusions.csv.

Supplementary Note S1. Revision analyses: designs and detailed results

Reference numbers refer to the main text. All analyses follow the pre-registration revision/PREREGISTRATION_REVISION.md (SHA-256 recorded 25 October 2026, 14:10 UTC, before any revision run) and its Amendment 1, which moved the runs to the original machine before any result was inspected.

S1.1 Pre-registration and platform

In response to review we ran six further analyses. We fixed them in a pre-registration hashed with SHA-256 before any of them ran (file revision/PREREGISTRATION_REVISION.md, 25 October 2026); an amendment, recorded before any result, moved the runs to the original machine. The equivalence margins in this plan came after the primary results were known.

On the original machine, refitting all 1,500 selected configurations reproduced every published validation and test score exactly. On Linux with the same package versions, the default configurations reproduce exactly but the tuned ones do not, because XGBoost draws its row and column subsamples through C++ standard-library distributions that differ between compilers.

S1.2 Designs

(i) Equivalence: for each optimizer pair we computed the paired differences in the primary test metric, their 95% confidence intervals and two one-sided t tests (TOST) with margins of 2% of the default test RMSE or 0.01 ROC-AUC, Holm-corrected over the 10 pairs of a cell. We call optimizers equivalent only when TOST rejects after correction. (ii) Pre-refit diagnostic: we refitted each selected configuration on the training set with early stopping, as during the search, and let this single model score both the validation and the test set. (iii) Validation control: we scored a pool of 200 random Experiment A configurations, the same for every dataset, by the single validation split and by 5-fold scaffold cross-validation on training plus validation data, with the test set untouched, and selected the best configuration by each criterion in 2,000 simulated random searches of 100 configurations. (iv) Descriptor-only mirror: Experiment B on BBBP and FreeSolv with only the descriptor genes mirrored, 30 seeds per optimizer. (v) Penalty and budget: Experiment B on BBBP and FreeSolv with seeds 0–9 under a zero penalty, under penalties of 1% and 5% of the default validation metric, and with a budget of 300 calls. (vi) Independent partitions: Experiment A on ESOL, FreeSolv, BBBP and BACE under five randomized balanced scaffold partitions, with descriptors re-filtered on each training set and three tuning seeds per optimizer. Partition entered a two-way analysis as a random block, and we tested the optimizer effect against the optimizer-by-partition interaction.

S1.3 Default configuration and model randomness

The default configuration is XGBRegressor or XGBClassifier from XGBoost 3.4.1 with every hyperparameter at its library value: 100 trees, maximum depth 6, learning rate 0.3, subsample and colsample_bytree 1, min_child_weight 1, reg_lambda 1 and reg_alpha 0. It uses the histogram tree method, one thread, squared-error or logistic loss, all descriptors and the library's default random state, with no early stopping and no evaluation set. Its validation score comes from a model fitted on the training set and its test score from a model refitted on training plus validation data, as for the tuned configurations. It samples neither rows nor columns, so its predictions do not depend on any seed.

The tuned models pass the tuning seed to XGBoost as the random state in every fit of a run, during the search and in the refit. Row and column subsampling therefore follow the same random stream within a run and a different stream in each run, so the spread across seeds contains both search and subsampling variation. Refitting a configuration with the same seed reproduces its fitness exactly (one thread, deterministic histogram construction), so the cache saves time without changing any fitness value. To separate the two sources, we refitted every selected configuration with two further model seeds (1000 and 1001).

S1.4 Statistical implementation

All tests are two-sided. Wilcoxon tests use Pratt's treatment of zero differences, which keeps zeros in the ranking but drops them from the signed sum; zeros occurred in 22 of the 3,000 paired optimizer comparisons, where two searches selected the same configuration, and in 2 of the 1,500 comparisons with the default. Tied values receive average ranks, and p values then come from the normal approximation. We ran the pairwise tests in every cell without an omnibus gate. A Holm family consists of the 10 pairs of one dataset and experiment, or of the five tests against the default; each secondary metric forms its own families. Bootstrap intervals resample the 30 seeds (10,000 percentile resamples) and keep the validation and test scores of a seed together for Spearman correlations. The interval for Nogueira stability is the asymptotic interval of [33]. Holm control holds within a cell, so across the 10 cells the chance of at least one false rejection under a global null can reach about 0.40.

S1.5 Optimizer comparison, tuning verdicts and independent partitions

A non-significant test does not show that optimizers are interchangeable. At the pre-specified margins, 30 of the 100 optimizer pairs were equivalent after Holm correction (Table S11): all 20 on Lipophilicity, 4 and 0 on ESOL in Experiments A and B, 3 and 2 on BBBP, 1 and 0 on BACE, and none on FreeSolv. On FreeSolv the 90% intervals allow differences of up to 0.21 kcal/mol (7.6% of the default RMSE), and on ESOL in Experiment B up to 0.061 log units, so these benchmarks cannot exclude practically relevant differences. Neither of the two significant pairs survived a Benjamini–Hochberg correction over all 100 pairs (smallest $q = 0.10$), and no Friedman test survived a Holm correction over the 10 cells (smallest adjusted $p = 0.09$).

XGBoost's own subsampling produced much of the seed-to-seed spread. Refitting each selected configuration with three model seeds showed that the model seed accounted for a median of 36% of the test-score variance within a cell (range 11–71%; Table S12). Averaging the three refits reduced this noise and changed one verdict: on BBBP in Experiment A, GWO beat TPE, WOA and PSO after Holm correction, by median differences of 0.005–0.008 ROC-AUC, inside the 0.01 margin. The FreeSolv Experiment A difference remained (one Holm-significant pair), and the other eight cells showed none.

The optimizer comparison held on five independently constructed scaffold partitions (Table S13). The optimizer effect, tested against its interaction with the partition, was not significant on ESOL ($p = 0.51$), FreeSolv ($p = 0.63$), BBBP ($p = 0.17$) or BACE ($p = 0.39$), and no pair differed after Holm correction. The variance between partitions exceeded the variance between tuning seeds on all four datasets, so our seed-level intervals understate the uncertainty about new chemical series. With five partitions the intervals were too wide to show equivalence.

Whether tuning helps depended on the split as well as on the dataset. On the five new partitions, every optimizer beat the default on ESOL in all five partitions, by a mean of 0.058–0.079 log units, the reverse of the result on the original split. On FreeSolv the optimizers beat the default in three or four of five partitions, and on BBBP and BACE their mean improvement was negative for every optimizer (Table S13). The verdicts of this section therefore describe the original split rather than the datasets.

S1.6 Validation reliability

Refitting does not explain the weak link. When the same pre-refit model scored validation and test, 40 of the 50 correlations had intervals that included zero, against 46 of 50 for the refitted models (Fig. S7; Table S14). Refitting lowered the median correlation from 0.07 to 0.01, a median change of −0.02 (interquartile range −0.18 to 0.04). Before refitting, validation tracked test on FreeSolv ({i}ρ{/i} = 0.42–0.52 in five cells) and in two Lipophilicity cells, while three cells showed significant negative correlations. The default configuration measures distribution shift without selection: its pre-refit test score sat 0.08 RMSE above its validation score on ESOL, 0.82 below on FreeSolv and 0.05 ROC-AUC below on BACE. Relative to these shifts, selection added optimism of 0.16–0.18 RMSE on ESOL, 1.1–1.2 kcal/mol on FreeSolv, 0.04 ROC-AUC on BBBP and 0.03 on BACE, and none on Lipophilicity.

Range restriction explains part of the pattern. Within a search the validation scores spread widely, but the scores selected by the 30 runs covered about half the 5–95% range of all searched scores on Lipophilicity (ratio 0.44–0.52) and 0.70 of it on ESOL (Table S15). The configuration pool tested this directly (Fig. S8; Table S16). Across all 200 configurations, the single validation split ranked test performance on Lipophilicity ({i}ρ{/i} = 0.81), BBBP (0.46) and BACE (0.35), but not on ESOL (0.03) or FreeSolv (−0.13). Among the 20 best configurations the correlation fell to between −0.39 and 0.14. Our pre-registered prediction that this drop would be significant on at least three datasets failed: it was significant on Lipophilicity and BACE only.

Five-fold scaffold cross-validation ranked configurations better than the single split on ESOL ({i}ρ{/i} = 0.54 against 0.12), FreeSolv (0.32 against 0.06) and Lipophilicity (0.80 against 0.74), with no clear change on BBBP or BACE. In the simulated searches, selection by cross-validation lowered the regret on ESOL from 0.096 to 0.038 RMSE and on FreeSolv from 0.54 to 0.31 kcal/mol, and the share of searches that beat the default rose from 1% to 88% on ESOL and from 87% to 100% on FreeSolv. On Lipophilicity, BBBP and BACE cross-validation did not help; on BBBP the share fell from 50% to 13%.

The configuration pool supports this reading: across 200 configurations the validation split ranked test performance on the three larger datasets, but not among the 20 best (Section S1.5).

The independent partitions point to the second mechanism. On ESOL, tuning lost to the default on the original split but beat it on all five new partitions, so a single split can decide whether tuning appears to help. Inner cross-validation, which averages over several groups of scaffolds, turned the ESOL verdict as well: selection by five-fold scaffold cross-validation beat the default in 88% of simulated searches, against 1% for the single split. On small datasets, the choice of validation scheme mattered more than the choice of optimizer.

S1.7 Mirrored encodings and prediction-level stability

Paired by seed, mirroring raised GWO's selected fraction by 0.112 on BBBP (95% CI 0.091–0.133) and by 0.116 on FreeSolv (0.092–0.140; Table S17). Its test ROC-AUC on BBBP changed by −0.002 (−0.010 to 0.006), but the equivalence test at 0.01 did not reject after Holm correction (p = 0.06), and its FreeSolv test RMSE changed by −0.02 kcal/mol (−0.14 to 0.10), an interval much wider than the 0.056 margin. The mirror thus produced no detectable change in accuracy, but the data do not establish that compact subsets cost nothing.

Mirroring only the descriptor genes, with the eight hyperparameter coordinates and their part of the shared start unchanged, reproduced the full effect (Fig. 5 of the main text). GWO's median fraction rose from 0.440 to 0.560 on BBBP (higher in 30 of 30 seeds) and from 0.430 to 0.553 on FreeSolv (29 of 30), Holm {i}p{/i} <

0.001 in both, while random search did not move (Holm {i}p{/i} = 0.35 and 1.00). The descriptor-only and full mirrors differed by 0.004 on BBBP (95% CI −0.015 to 0.024) and 0.005 on FreeSolv (−0.019 to 0.028); TOST at ±0.03 gave {i}p{/i} = 0.005 and 0.016, or 0.021 and 0.053 after Holm correction over the five optimizers. Reversing the hyperparameter coordinates thus contributed nothing we could measure, and the descriptor genes carry the bias.

Different columns did not mean different predictions. The test predictions of two runs of the same method correlated with a mean Pearson {i}r{/i} of 0.89–0.98 across cells in Experiment B, against 0.91–0.99 in Experiment A, where every run used all descriptors (Table S18). The many correlated RDKit descriptors let different subsets carry nearly the same information.

S1.8 Penalty, budget, compute and chemical dependence

The subset-size penalty did not drive the results (Table S19). With seeds 0–9 on BBBP and FreeSolv, no Friedman test reached significance and no pair differed after Holm correction under a zero penalty or under penalties of 1% and 5% of the default validation metric (weights from 0.0008 on BBBP to 0.16 on FreeSolv). Kendall's {i}τ{/i} between the mean-rank order of each condition and the reference ranged from 0.4 to 1.0. GWO's median fraction stayed at 0.42–0.43 in every condition, and without any penalty the mirror still raised it, from 0.42 to 0.56 on BBBP (10 of 10 seeds, {i}p{/i} = 0.002) and from 0.43 to 0.52 on FreeSolv (9 of 10, {i}p{/i} = 0.010), while random search did not change.

Tripling the budget to 300 calls lowered the mean best validation fitness of every optimizer, significantly after Holm correction for TPE and PSO on BBBP and for TPE, GWO and PSO on FreeSolv, but no optimizer's test score changed significantly (smallest Holm {i}p{/i} = 0.21; Table S20). The optimizers still did not differ (Friedman {i}p{/i} = 0.11 on BBBP and 0.40 on FreeSolv), subset stability stayed at or below 0.058, and GWO selected even fewer descriptors (0.42 to 0.38 on BBBP, 0.43 to 0.36 on FreeSolv), at 2.1–3.8 times the compute. The unfinished convergence in Fig. S6 therefore does not explain the null result: longer searches fitted the validation set better without improving test performance.

Elapsed compute tracked the call count. Fitting time grew in proportion to the number of calls, so a quarter of the compute corresponded to a quarter of the budget for every method (Table S10). Random search, TPE, GWO and PSO evaluated 100 unique configurations in every run, and WOA a median of 97 (range 63–100). In each dataset and experiment, 17–31% of the runs still improved their validation fitness in the last ten calls.

Chemical dependence across splits changed little. After standardization (largest fragment, neutralization, canonical tautomer, stereochemistry removed), parent structures crossed splits only in BBBP (3 groups) and BACE (12 groups). Excluding the affected validation and test molecules and those in groups with conflicting labels (2 and 1 in BBBP, 3 and 9 in BACE) left every optimizer comparison and every tuning verdict unchanged, except on BACE in Experiment B, where the Friedman {i}p{/i} fell from 0.18 to 0.047 without any Holm-significant pair and PSO now beat the default (Table S21). The original protocol remains the reference analysis.

S1.9 Relation to out-of-distribution benchmarks

Chemically informed out-of-distribution benchmarks, such as the Lo-Hi splits [38] and the property-range splits of BOOM [39], measure how far models extrapolate. We held the learner, representation, split and budget fixed and asked only whether the optimizer changes the answer. Harder splits would widen the validation–test shift that limits tuning, and we expect them to make the optimizer matter even less, a prediction we have not tested.

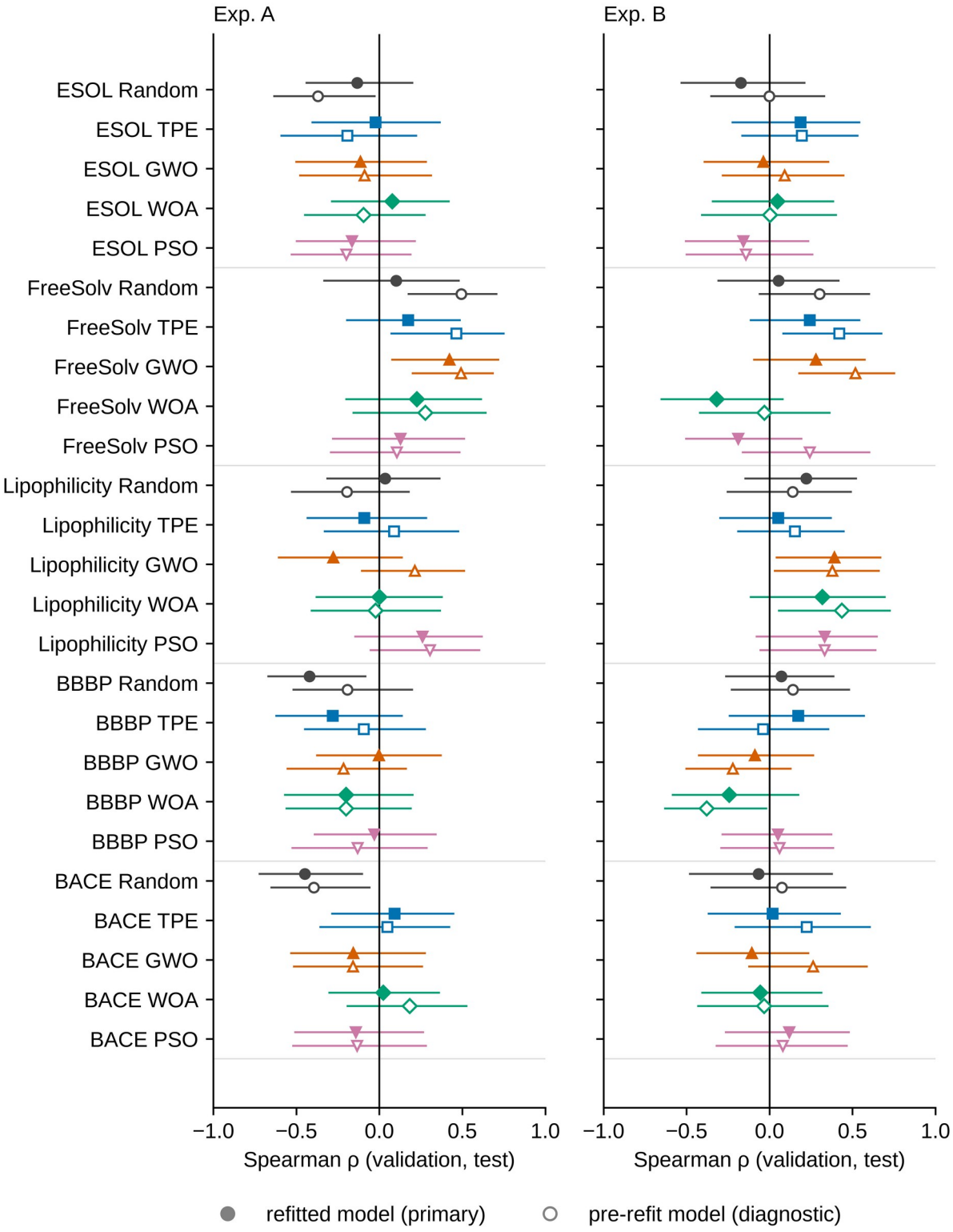


Figure S7. Spearman correlation between best validation and test scores across the 30 seeds, per dataset, method and experiment, with bootstrap 95% confidence intervals. Filled markers: refitted models (primary analysis). Open markers: the same configurations before refitting, with one model scoring validation and test (diagnostic).

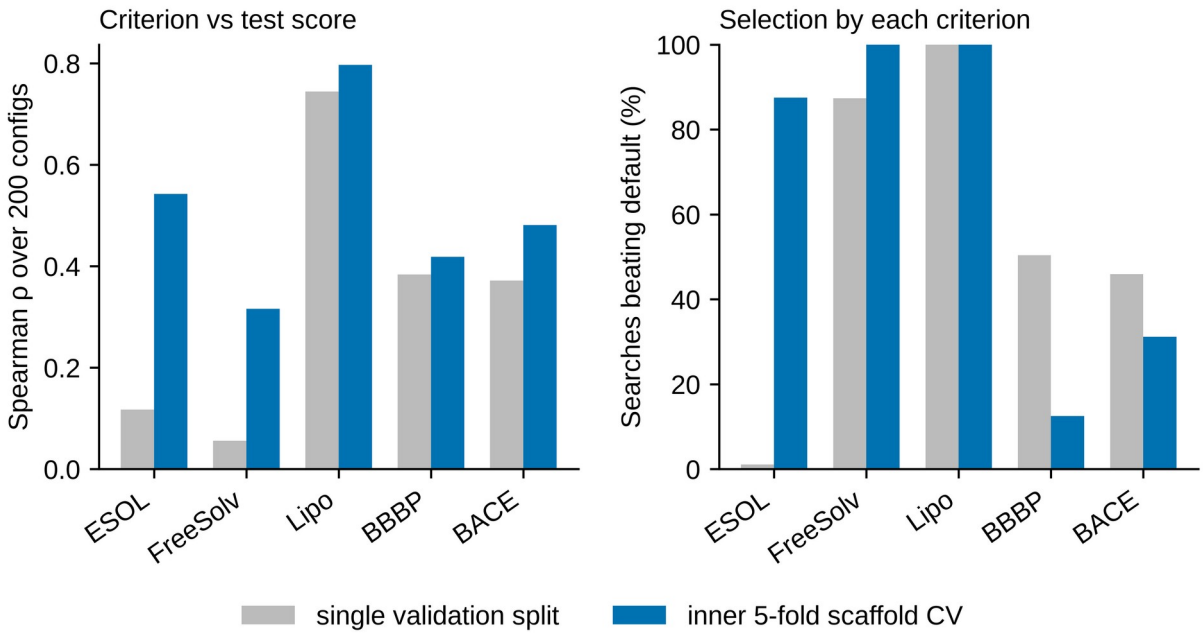


Figure S8. Validation control on a pool of 200 random Experiment A configurations. Left: Spearman correlation between each criterion and the refitted test score across the pool. Right: share of 2,000 simulated random searches of 100 configurations whose selected configuration beat the default on test.